RGB-Thermal Fusion for Improved Object Detection

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Introduction

Object detection plays a crucial role in a variety of applications such as environmental sensing, autonomous driving, search-and-rescue, and robotics. Normally, object detection techniques use standard red green blue (RGB) cameras, which suffer in challenging environmental conditions like glare, low-light, or inclement weather.

By measuring infrared radiation, thermal cameras can provide valuable complementary information to improve overall accuracy. Sensor fusion describes the broad category of techniques used to combine information from disparate sensing sources. Within the field of object detection, convolutional neural networks (CNNs) currently offer the highest performance.

Dataset

In order to perform multispectral sensor fusion, 1:1 RGB to thermal frames were needed with sufficient class labels. The vehicle perspective offered the greatest number of potential candidates, and the Teledyne FLIR ADAS Thermal Dataset was ultimately selected [1].

Details:

- Recorded from vehicle perspective.
- 15 principal class types (including car, person, bike, truck, light, …).
- Total of 3,749 matched RGB : thermal frames with manually annotated bounding boxes.

Shown left is an RGB image from the training dataset, displaying all ground-truth bounding boxes with class labels. Shown right is the class label breakdown.

Methodology

The project began with a state-of-the art single-stage object detection model, You Only Look Once (YOLO) v7 [2].

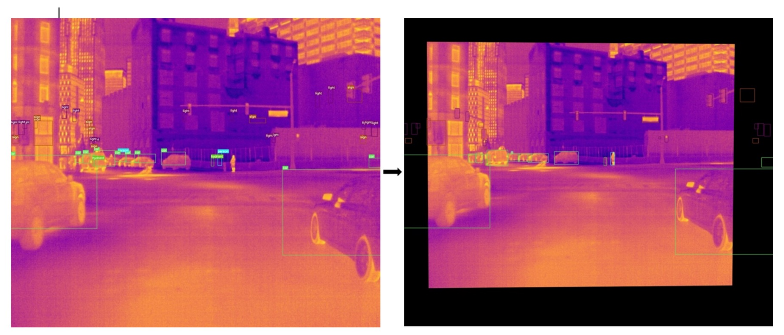
Baseline:

- With augmentation, trained RGB-only model and thermal-only models with batch sizes of 16 and 32 at 100, 200, and 300 epochs.

Shown above is a matched image pair with concatenation.

Early fusion:

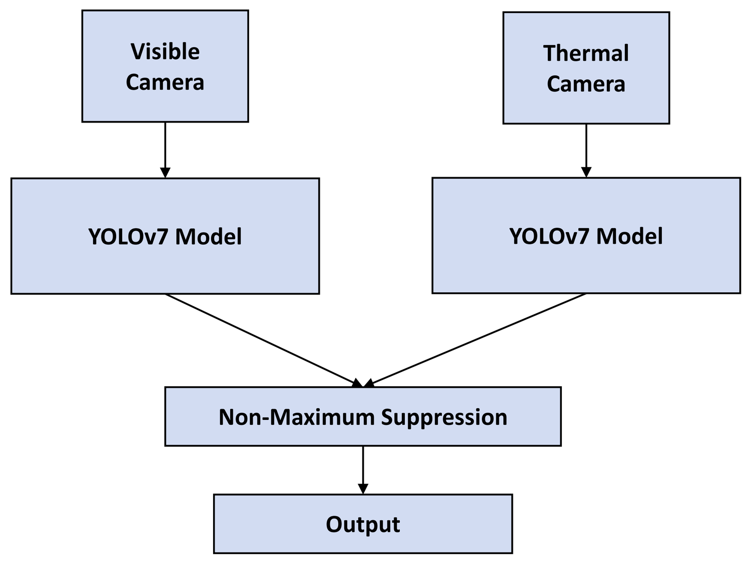
- Applied planar homography to IR frames based on manually-annotated corresponding points for 1224x1024 and 1024x768 images, retaining RGB labels.
- Modified input layer of YOLOv7 to accept 6-channel input with a custom dataloader.
- Trained with batch sizes of 16 and 32 at 100, 200, and 300 epochs.



Late fusion:

- Used separately trained RGB and thermal-only networks to perform inference separately then concatenated results
- Modified the Non-Maximum Suppression (NMS) step in the detection pipeline with Soft-NMS [3] and/or a merge step for bounding box curation

Shown right is a diagram of the late fusion approach with pre-trained individual networks. First ranking bounding boxes by confidence, NMS removes overlapping boxes with a hard threshold for Intersection over Union (IoU). By contrast, Soft-NMS instead decays confidence scores by the boxes IoU. The merge step combines boxes by using a weighted mean of the scores.



Results

At 100 and 200 epochs, the early fusion models significantly outperformed the RGB and thermal-only models in terms of the target metric, mean Average Precision with an IoU threshold of .5. At 300 epochs, this advantage narrowed and was surpassed by the thermal models.

Model	P	R	mAP@.5	mAP@.5:.95	Latency (ms)
RGB-16	.837	.757	.765	.506	7.2
RGB-32	.830	.783	.770	.506	7.5
Th-16*	.850	.843	.836	.555	6.1
Th-32*	.837	.849	.847	.559	6.2
Merge 1-16	.855	.829	.832	.569	8.0
Merge 1-32	.927	.909	.909	.582	9.0

Shown above are Precision, Recall, mean Average Precision, and latency for the models trained at 200 epochs, dash indicates batch size. All training and testing took place on an NVIDIA V100 GPU.

However, the late fusion models performed worse than the baseline with a substantial latency hit. This latency was due to double inference and an inefficient implementation of Soft-NMS.

Merge	Soft	P	R	mAP@.5	mAP@.5:.95	Latency (ms)
False	False	.7	.771	.653	.365	13.7
True	False	.702	.774	.657	.385	13.2
False	True	.703	.754	.699	.395	140.2
True	True	.707	.759	.703	.428	139.1

Shown above are the same metrics with the merge step and Soft-NMS using RGB-16-200 and Th-16-300 weights.

Conclusions

The early fusion models performed much better than the baseline with only a minimal latency hit. In contrast, the late fusion ensemble approach performed worse than the baseline despite improvement from NMS modifications. Future work includes the expansion of testing to other datasets and integrating models into an autonomous system.

Acknowledgements

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[1] "Teledyne FLIR Free ADAS Thermal Dataset v2." Teledyne FLIR LLC, Jan. 2022. [Online]. Available: https://www.flir.com/oem/adas/adas-dataset-form/

[2] C.-Y. Wang, A. Bochkovskiy, and H.-Y. M. Liao, "YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors," 2022, doi: 10.48550/ARXIV.2207.02696.

[3] Q. Wang, Y. Chi, T. Shen, J. Song, Z. Zhang, and Y. Zhu, "Improving RGB-Infrared Object Detection by Reducing Cross-Modality Redundancy," Remote Sensing, vol. 14, no. 9, p. 2020, Apr. 2022, doi: 10.3390/rs14092020.